Assignment 02



BEAT- UP MECHANISM (WEAVING)

Prof: Anirban Guha

Name: Niraj Niraula ***Roll No.:*** - 24M1723

Department of Mechanical Engineering

Indian Institute of Technology Bombay

Powai, Mumbai, India

1. Existing mechanism

Beat-up, the final crucial step in weaving, involves firmly positioning the last weft pick against the previous ones in the fabric. This action is achieved by swiftly driving the reed, fixed within the sley, towards the clothes fell using the cranks on the crankshaft. As the sleywood moves between shuttle boxes, the shuttle travels along its race during its backward motion. Beat-up, the third primary motion in weaving, marks the final step of the process on a loom. It involves firmly positioning the last inserted weft pick against the preceding ones in the fabric. This is achieved by driving the last pick of weft towards the fabric's fell, facilitated by a reed fixed in the sley. The sley undergoes a sudden and swift movement towards the fabric's fell, propelled by the cranks in the crankshaft. As the sleywood moves between shuttle boxes, the shuttle traverses over its race during its backward motion.

1.1 Images

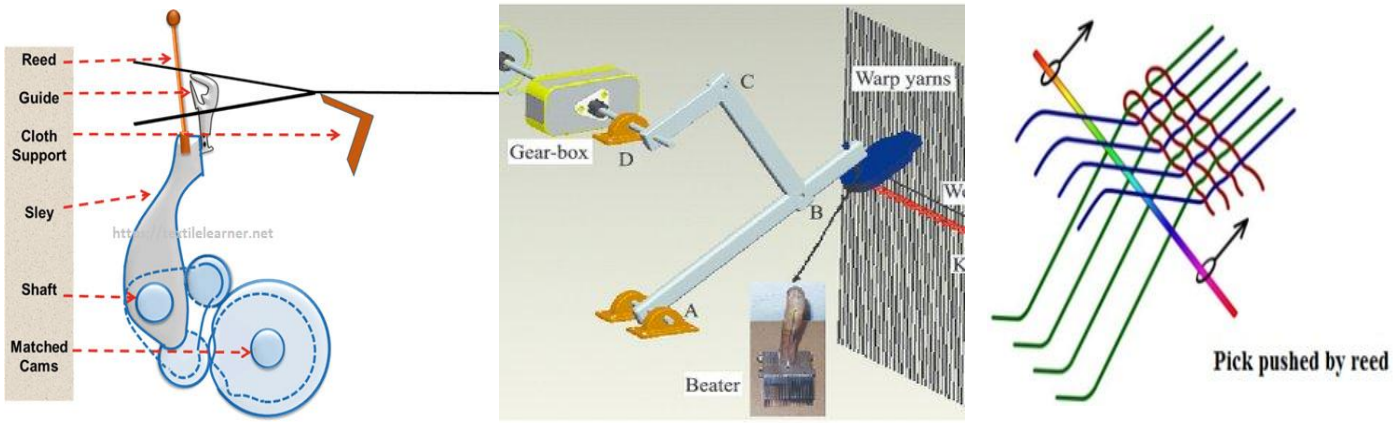


Fig.1 Images of original mechanism

1.2 Description

Beat up is the third primary motion in weaving. It consists in driving the last pick of weft to the fell of the cloth. This is accomplished with the help of a reed fixed in sley. The sley is given a sudden and quick movement towards the fell of the cloth by the cranks in the crankshaft. The sleywood runs from one shuttle box to another, and when at its backwards movement, the shuttle travels over its race. The objective of the beat-up mechanism is to push the newly inserted pick to the cloth fell(fell is the last position of the last pick in the cloth woven on the (loom) and to ensure uniform pick spacing. In weaving, various beat-up mechanisms exist,

typically categorized by their operating elements. Generally, two primary types are recognized based on this distinction. One is the crank and crank-arm beat-up, predominantly utilized in shuttle looms, while the other is the cam beat-up, more commonly found in shuttle less looms. Despite this differentiation, the essential components of the beating process remain consistent, regardless of whether the motion originates from a crank or a cam. The fundamental disparity lies in the source of motion for the slay, which ultimately dictates the overall parameters of the beat-up process. write in simple language

Beat-up in Weaving: A Simplified Explanation

Beat-up is a crucial step in weaving where the last thread of yarn (weft) is pushed to the edge of the fabric being woven. This is done using a tool called a reed, which is attached to a part of the loom called the sley. The sley moves quickly towards the fabric edge, driven by cranks or cams, depending on the type of loom.

Purpose of Beat-up:

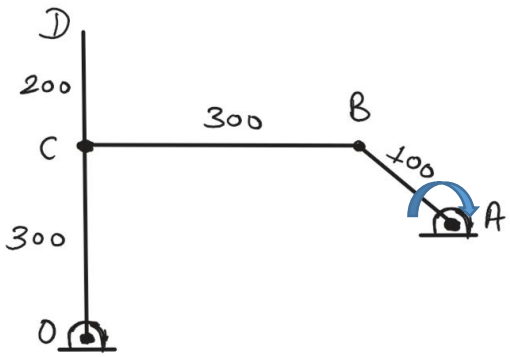
- **Push the Weft:** Move the newly inserted yarn thread to the edge of the fabric.
- **Even Spacing:** Ensure that the yarn threads are evenly spaced in the fabric.

Types of Beat-up Mechanisms:

1. **Crank and Crank-arm:** Commonly used in traditional looms with shuttles.
2. **Cam Beat-up:** Often used in modern looms without shuttles.

Both methods achieve the same goal but differ in how the motion is created.

2 Kinematic Sketch



AB=crank, BC=coupler, OD=rocker

Fig.3 kinematic sketch of original mechanism

3 Simulation of Original Mechanism in Software

The dimensions of the mechanism are determined according to the specifications of the machine and the desired output at the focal point. Additionally, angular and linear speeds are estimated based on the range of operating conditions. These are the isometric and front view of the approximate mechanisms modeled in SOLIDWORKS software and analysis is done in ADAMS.

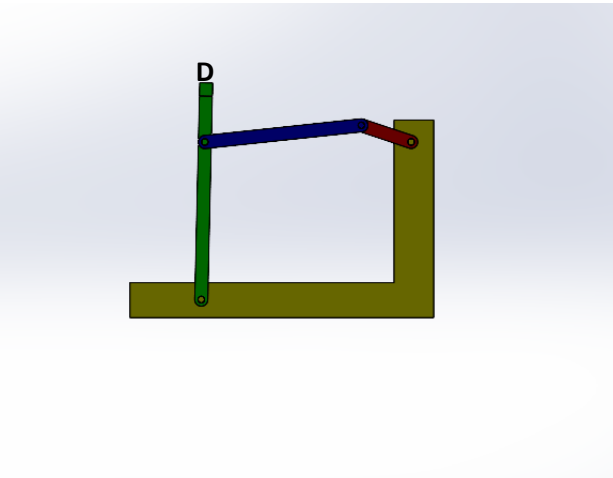


Fig.4 front view of original mechanism

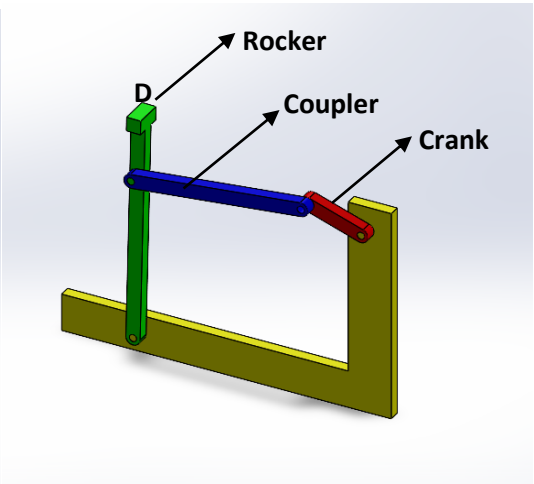


Fig.5 isometric view of original mechanism

Link Number	Link name	Link length(mm)
Link 1	Crank(input)	100
Link 2	Connecting link(coupler)	300
Link 3	Rocker (sley sword)	500
Link 4	Ground	NA

Connection	Type of link
Link 1 and link 2	Revolute
Link 2 and link 3	Revolute
Ground and link 1	Revolute
Ground and link 3	Revolute

4.1 Displacement Plots

4.1.1 Displacement

The mechanism was analyzed for displacement on the considered point D of rocker as shown in the fig.3 and following are the plots of x & y components of the same link.

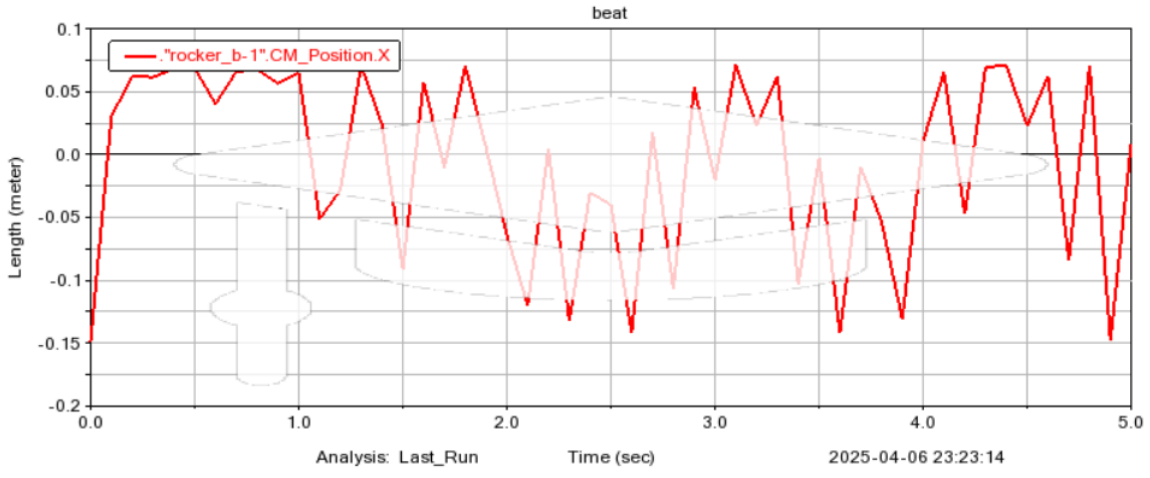


Fig.6 X-axis Displacement plot of original mechanism

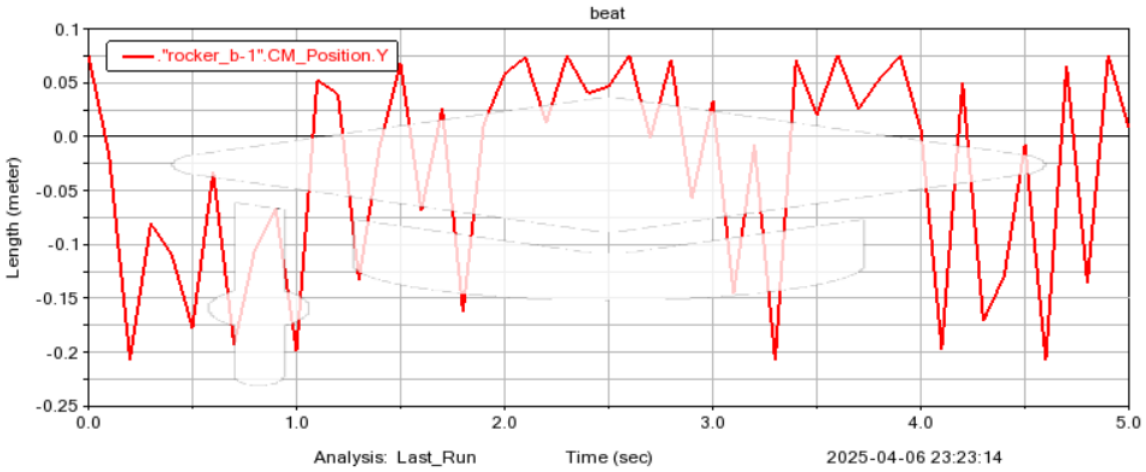


Fig.7 Y-axis Displacement plot of original mechanism

4.1.2 Trace

Also the point of interest on the output link was analyzed for the path its traces over a cycles of motion.

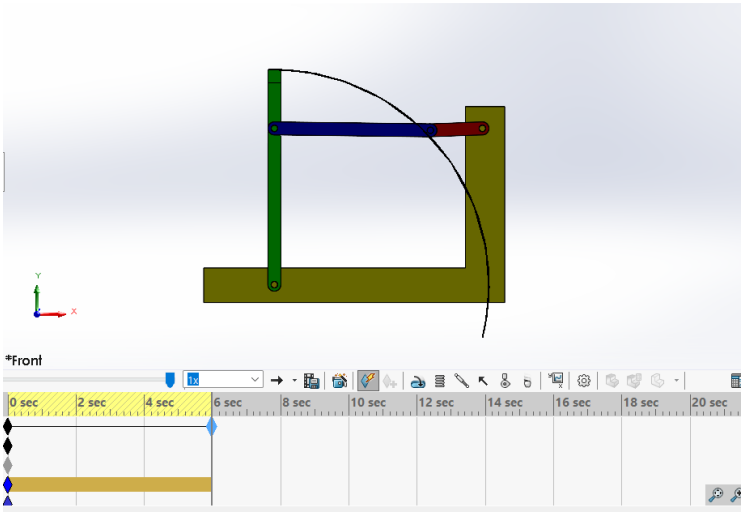


Fig.8 Trace of original mechanism

4.2 Velocity

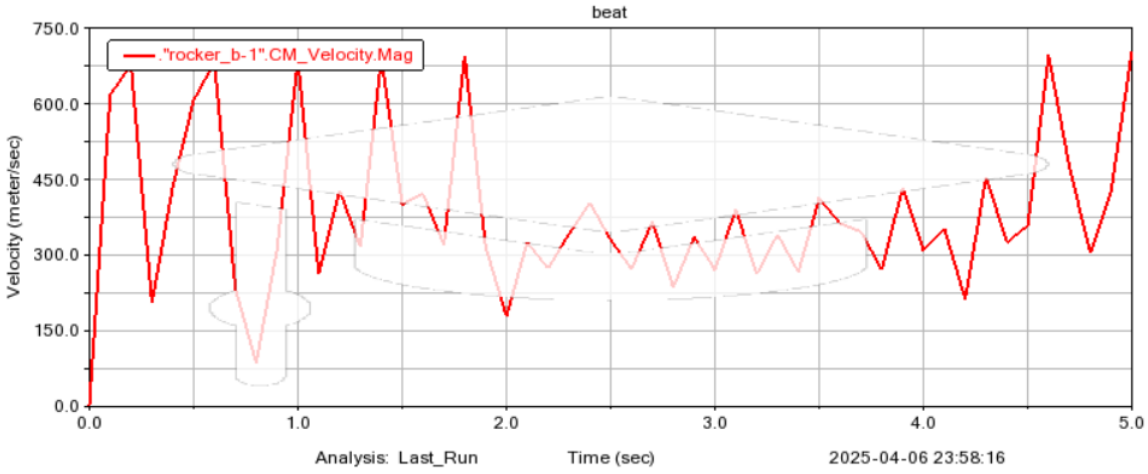


Fig.10 velocity of original mechanism

4.3 Acceleration

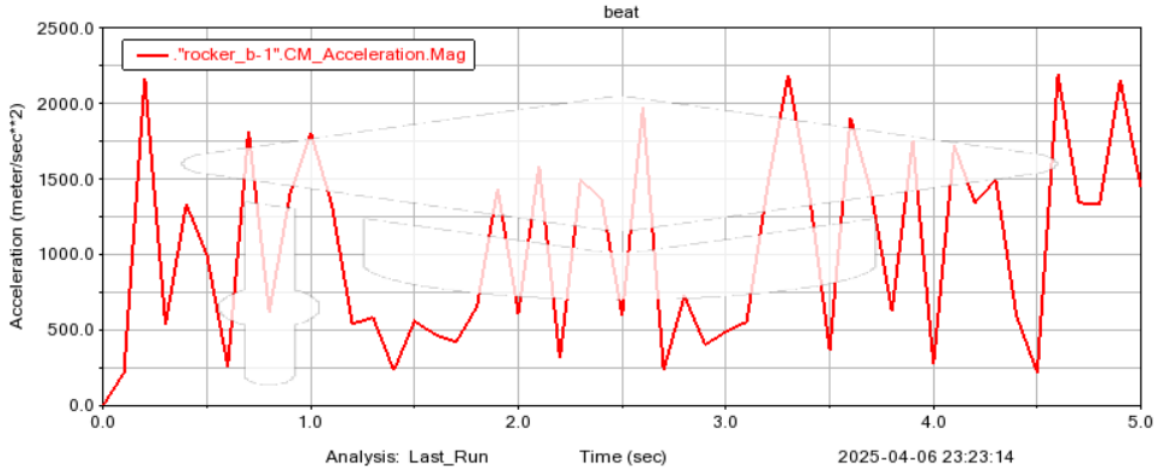


Fig.11 Acceleration of original mechanism

5 Dynamic Analysis of Original Mechanism: Simulation in Software

5.1 Total Mass

Link Name	Mass (KG)
crank	0.428
coupler rod	1.227
Rocker	1.8

The total mass of the mechanism is 3.445

5.2 Torque/Force

The methodology adopted to find out input link torque was by doing dynamic analysis by applying force at output link and then measuring the torque generated at the input link by the motor.

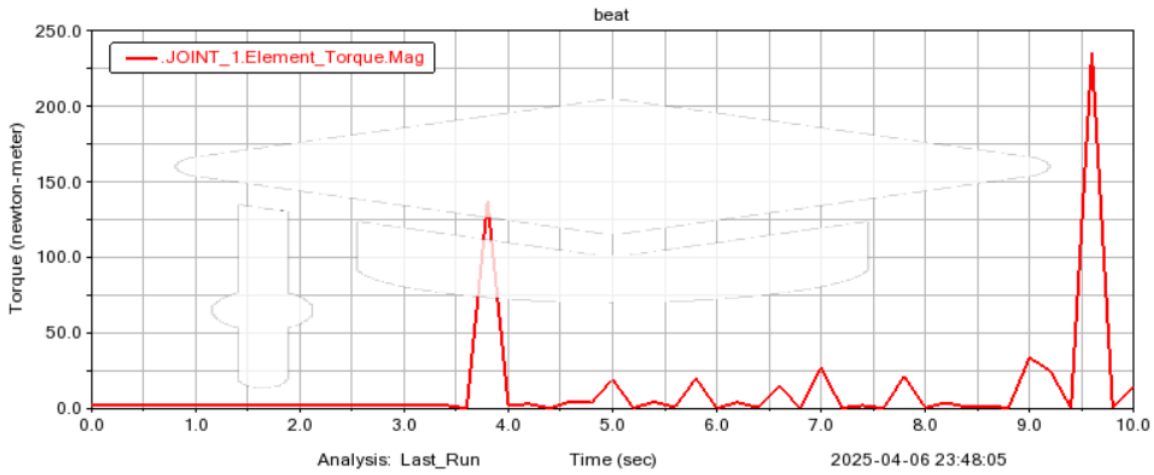


Fig.11 Torque of original mechanism

5.3 Power

The power required by the input motor can be easily obtained by considering the average motor torque(25 N-m) required and then multiplying by the constant angular velocity of the input arm which comes out to be

Speed of the input arm N = 150 rpm

Power= T * ω= 392.7 Watt

6 Alternate Mechanism

An attempt was made to devise a new linkage that would perform the same task and to get similar output .Images attached below is the kinematic linkage sketch of the proposed alternate mechanism with only lower pairs showing different links. Here my crank is an input and slider is the output the out on the alternate gives same horizontal straight line motion as my original mechanism.

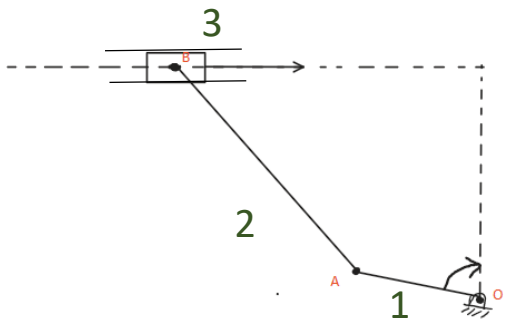


Fig.12 Kinematic sketch of Alternate mechanism

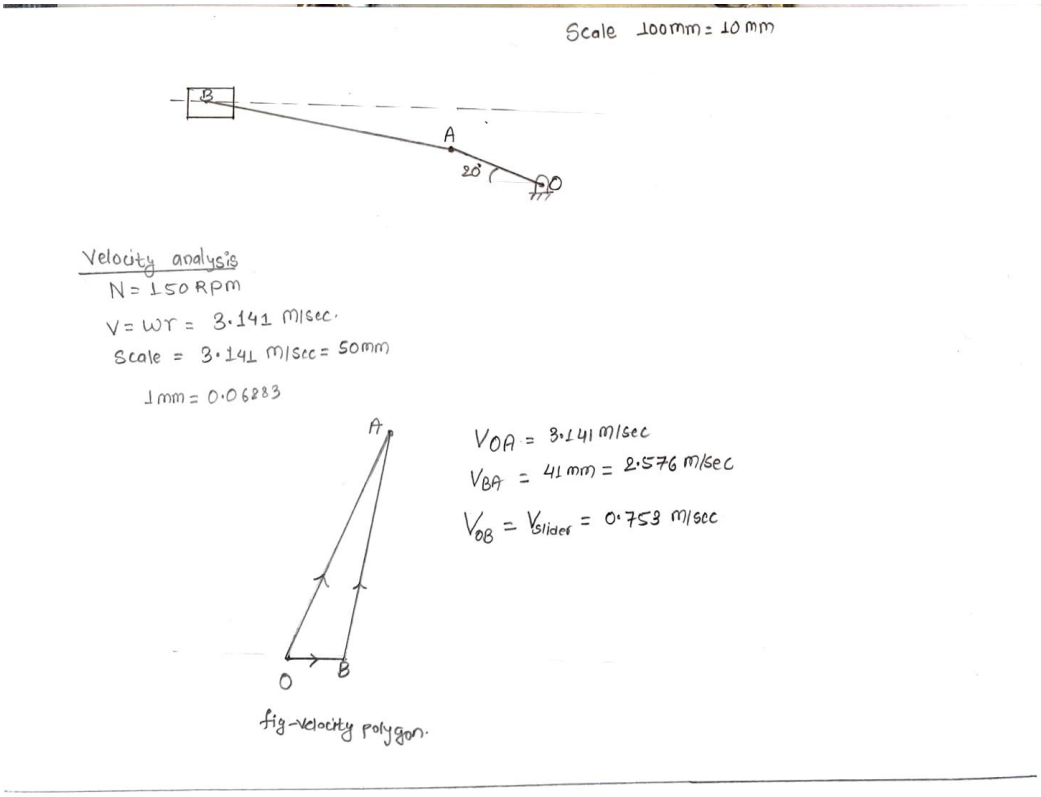
All the information about the mechanism linkage and their connection are given below:

Link Number	Link name	Link length(mm)
Link 1	Crank (OA)	200
Link 2	Connecting rod(AB)	500
Link 3	Slider	NA

Related links	Types of joint
Link 1 and ground	Revolute
Link and link 2	Revolute
Link 2 and link 3	Revolute
Link 3 and ground	Translation

7.1 Velocity Analysis:

Velocity analysis is done by graphical method taking same input 150 rpm as original mechanism in the orientation where crank angle is at 20°



Thus the velocity of slider is found to be 0.753 m/sec

7.2 Acceleration Analysis:

acceleration analysis. Scale = 49.32 = 50mm

$a_{OA}^{centripetal} = \frac{V_{OA}^2}{r_{OA}} = \frac{3.141^2}{0.2} = 49.32\ m/sec^2 = 50mm$

$a_{BA} = \frac{V_{BA}^2}{r_{BA}} = \frac{2.576^2}{0.5} = 13.271\ m/sec^2 = 13\ mm$

$a_{OB} = a_{slider} = 62mm = 62.85\ m/sec^2$

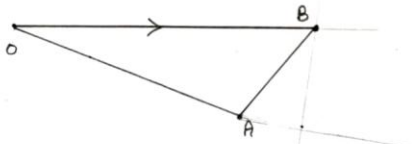


fig- Acceleration polygon

Thus the acceleration of the slider is found to be 62.85 m/sec^2

8.1 Modeling the alternate mechanism

8.1.1

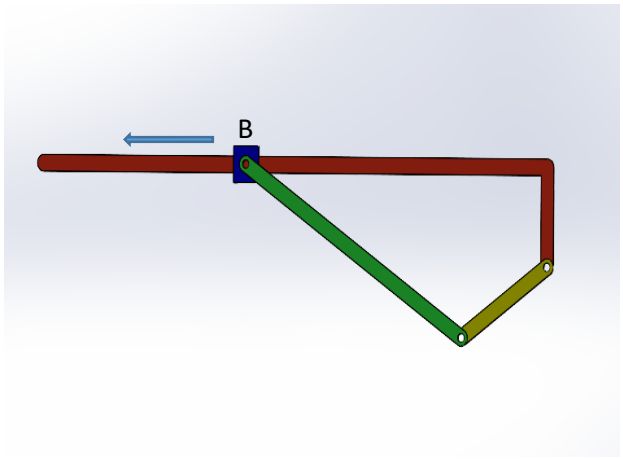


Fig.13 front view of Alternate mechanism

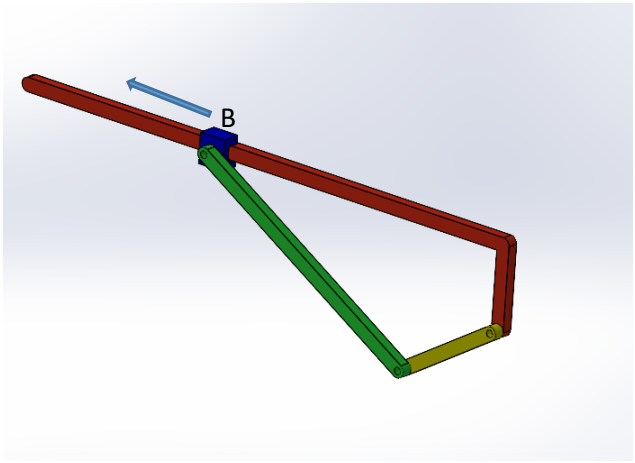


Fig.14 Isometric view of Alternate mechanism

8.1.2

Here slider is my required output and point B is my point of calculation.

Kinematic Simulation is done for rotational speed of 150 rpm and graph are extracted as shown below:

8.2 Displacement:



Fig.15 displacement X-axis curve of Alternate mechanism

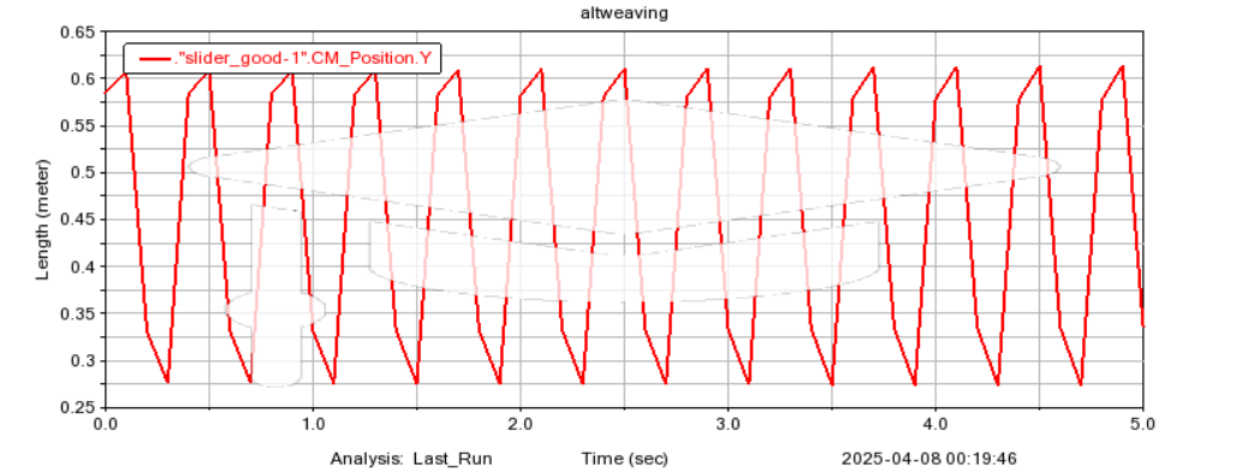


Fig.16 displacement Y-axis curve of Alternate mechanism

8.3 Trace:

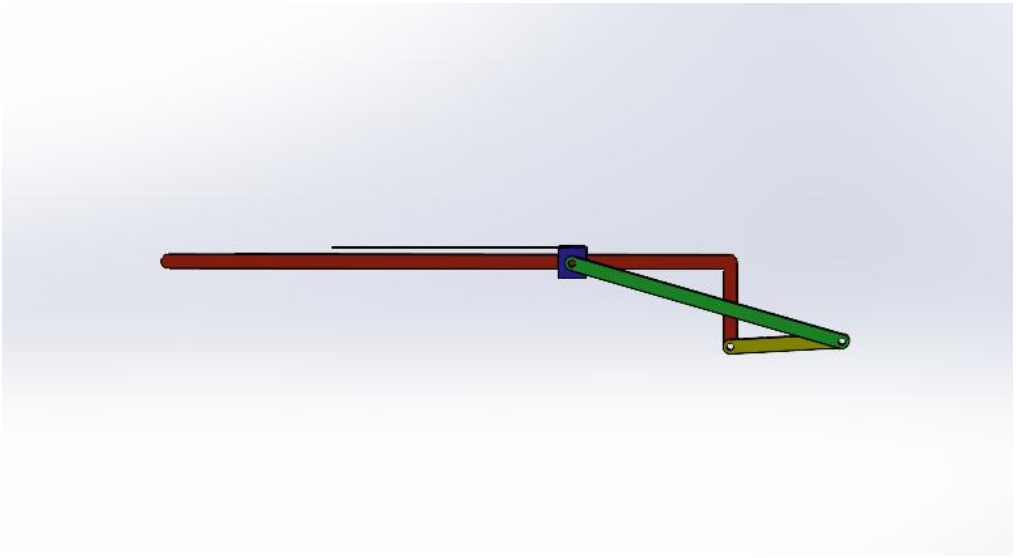


Fig.17 Trace curve of Alternate mechanism

9 Dynamic Analysis of Alternate Linkage: Simulation in Software

9.1 Total Mass:

Link Name	Mass (KG)
crank	0.81
connecting rod	1.988
slider	0.528

The total mass of the mechanism is 3.326 kg.

9.2 Torque/Force:

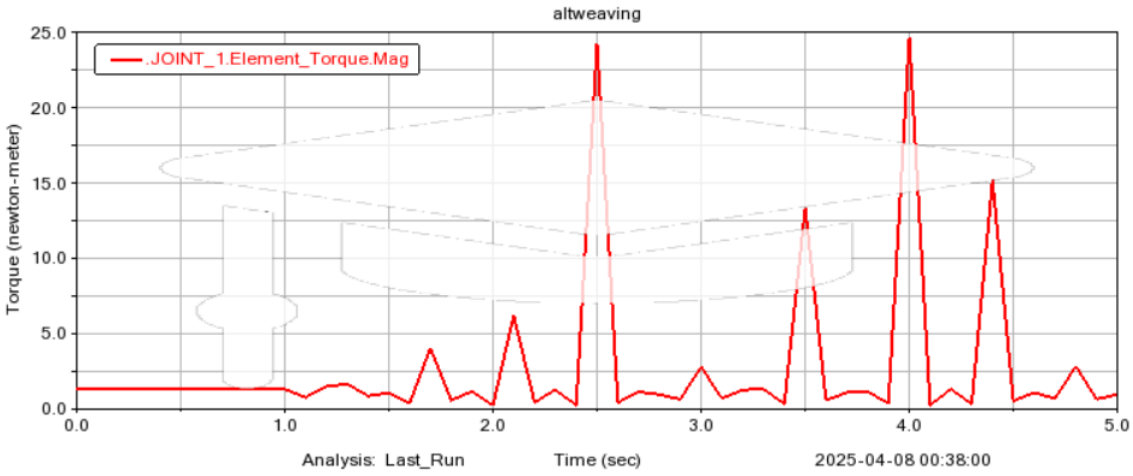


Fig.18 Torque curve of Alternate mechanism

9.3 Power:

The power required by the input motor can be easily obtained by considering the average motor torque(10 N-m) required and then multiplying by the constant angular velocity of the input arm which comes out to be
Speed of the input arm N = 150 rpm
Power= T * ω= 157.0 Watt

10. Comparison of Original and Alternate Mechanism

10.1 Displacement Comparison:

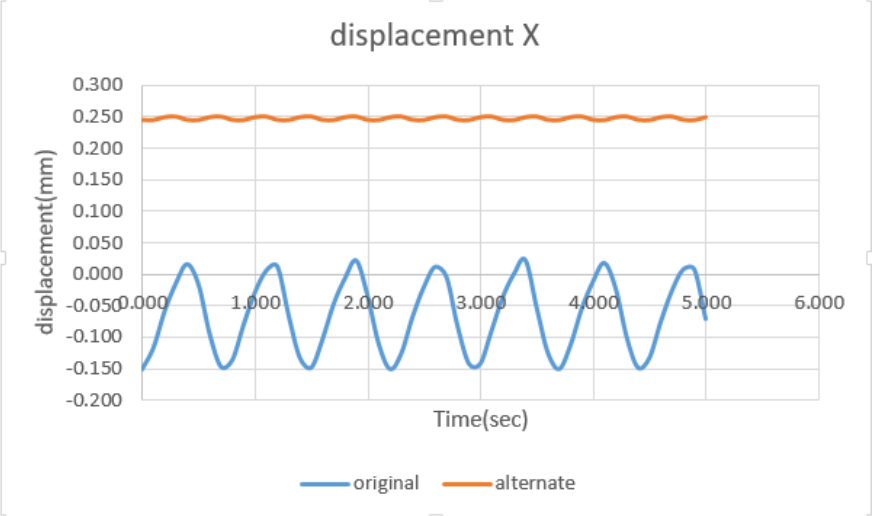


Fig.19 displacement X-axis comparison between mechanisms

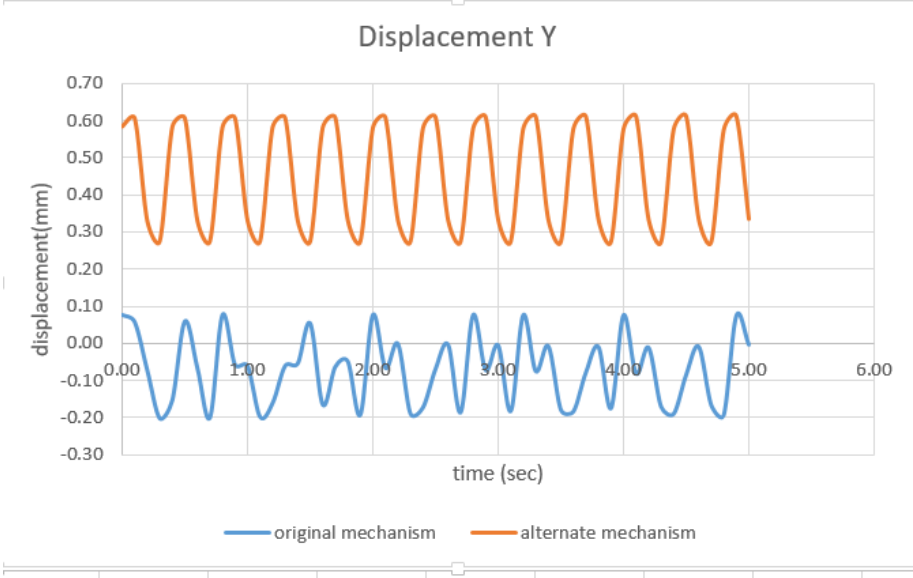


Fig.19 displacement Y-axis comparison between mechanisms

10.2 Trace Comparison:

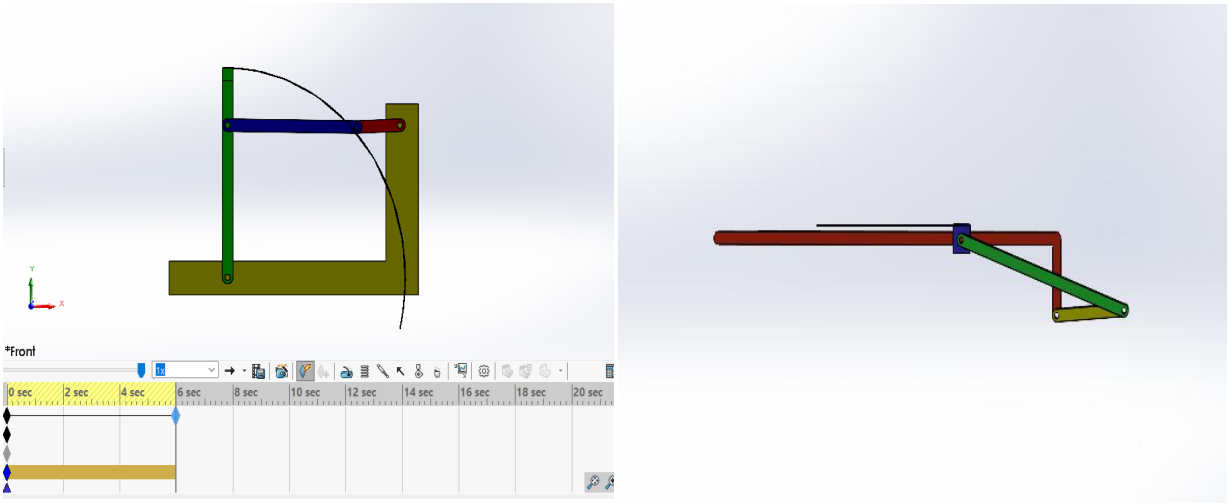


Fig.20 Trace comparison between mechanisms

Although the trace is different in both the mechanism the output function is same to push in the utilized mechanism.

10.3 Torque/Force Comparison:

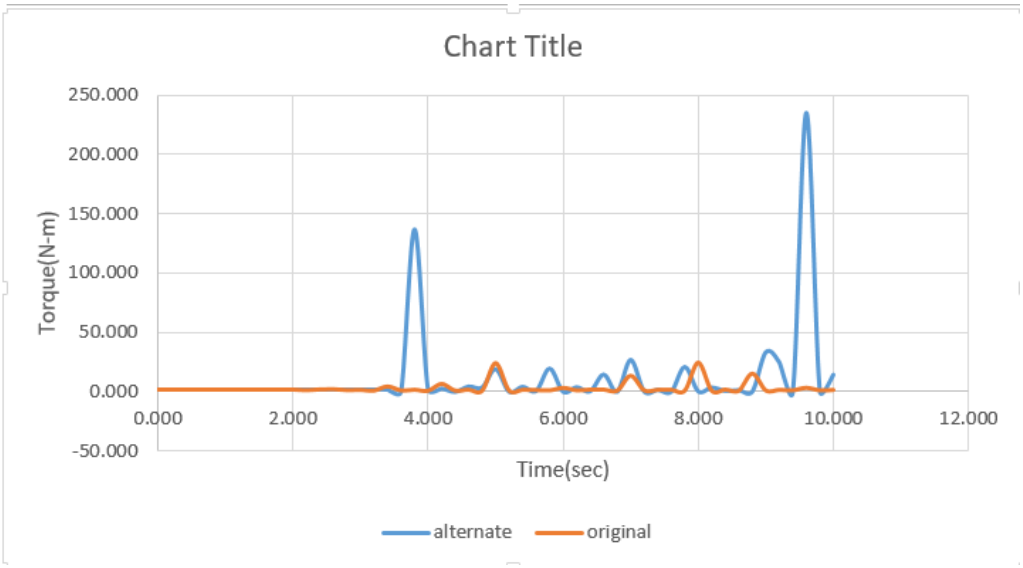


Fig.20 Torque comparison between mechanisms

The average value of torque of original mechanism is about double of then alternate mechanism as shown in the above graph.

10.4 Power Comparison:

Power consumption on the original mechanism is 392.7 watt whereas on the alternate mechanism was 157 watt. So the less power is required to run alternate mechanism than original mechanism.

11 Comments

11.1 Error Analysis:

1. Notably, Figure 18 highlights a substantial disparity in the x-component of displacement between the two mechanisms. While the existing mechanism demonstrates displacement in the x-direction, the alternate mechanism shows very low displacement.

2. In examining the Displacement plots, it's evident that the y-component pattern of the alternate mechanism not that much resembles that of the existing one, depicted in Figure 19. However, there's a notable similarity in the pattern.

3. This discrepancy primarily stems from the orientation of the mechanism/model within the Software. The most significant deviation in displacement within the alternate linkage results from the linear motion of the output link, a characteristic absent in the existing mechanism.

11.2 Index of Merit

● The alternative mechanism outperformed the current.
● Despite having similar masses, the alternate mechanism falls ahead in delivering the required motion only in the one axis as shown In the trace comparison.
● the alternate mechanism requires significantly less power compared to the existing one.
● Consequently, the existing mechanism boasts a superior index of merit.

References:

- https://www.google.com/url?sa=i&url=https%3A%2F%2Ftextilelearner.net%2Fbeat-up-mechanism-in-weaving-motion%2F&psig=AOvVaw0R8lKeFY8LpbvFJxWmsEXu&ust=1744177159608000&source=images&cd=vfe&opi=89978449&ved=0CBYQjhxqFwoTCNiB-bXcx4wDFQAAAAAdAAAAABAE
- https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.researchgate.net%2Ffigure%2FCrank-rocker-type-beat-up-mechanism-placed-in-front-of-the-handmade-carpet-loom-Fig_fig1_242741938&psig=AOvVaw0R8lKeFY8LpbvFJxWmsEXu&ust=1744177159608000&source=images&cd=vfe&opi=89978449&ved=0CBYQjhxqFwoTCNiB-bXcx4wDFQAAAAAdAAAAABAI
- https://www.google.com/url?sa=i&url=https%3A%2F%2Fin.pinterest.com%2Fpin%2F361625045087275384%2F&psig=AOvVaw0R8lKeFY8LpbvFJxWmsEXu&ust=1744177159608000&source=images&cd=vfe&opi=89978449&ved=0CBYQjhxqFwoTCNiB-bXcx4wDFQAAAAAdAAAAABAQ